Bayu Nismara Nagatama

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ABOUT

Motivated and detail-oriented Informatics Engineering student at Institut Teknologi Sepuluh Nopember (ITS) Surabaya, currently in my 7th semester, with a strong focus on front-end web development. I have hands-on professional experience through a full-time internship at Roundbytes and an ongoing contract role as a Front-End Developer at PLN Group, where I have built and maintained production-level web applications using modern frameworks including Vue.js 3, TypeScript, and Tailwind CSS. Beyond professional experience, I actively apply deep learning and AI concepts in academic research, including semantic segmentation on medical imaging data for my undergraduate thesis. I also have a track record in competitions, earning regional and international recognition in technology innovation and the arts. I thrive in collaborative, fast-paced environments and am eager to contribute meaningful, clean, and scalable front-end solutions while continuing to grow as a well-rounded software developer.

EDUCATION

Institut Teknologi Sepuluh Nopember (ITS) Aug 2023 — Present

Bachelor's Student of Informatics Engineering (Teknik Informatika) — GPA: 3.68 / 4.00

- Currently in 7th semester; active in academic projects spanning front-end web development, deep learning, and AI-assisted application design.
- Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Human-Computer Interaction, Machine Learning, Web Programming.
- Undergraduate thesis: Semantic segmentation of dental CBCT images using YOLO-based architecture to classify alveolar bone quality (Lekholm & Zarb D1–D4), with evaluation metrics DSC, IoU, HD95, Accuracy, Precision, Recall, and F1.

WORK EXPERIENCE

PLN Group — Front-End Web Developer (Contract) Oct 2025 — Present

PT PLN (Persero) — Surabaya, Indonesia

- Developed the front-end for LegalBot ID (Legal AI Suite), an internal AI-powered legal assistant platform for PLN comprising a Legal AI Assistant chatbot, a Document Analysis Tool for automated contract/regulation review, and a Legal Search Engine with semantic document understanding.
- Built the user interface using Vue.js 3, TypeScript, Tailwind CSS, and shadcn-vue, consuming a FastAPI backend integrated with a fine-tuned open-source LLM (Qwen2.5-7B-Instruct) specialized for Indonesian legal domain Q&A.
- Designed and built a premium, responsive web design for the LegalBot ID landing page and dashboard, ensuring cross-browser compatibility and mobile responsiveness.
- Collaborated closely with AI/ML and backend teams in an enterprise environment, translating system requirements and API contracts into polished front-end experiences while managing tasks across multiple internal tools.

Roundbytes — Full-Time Web Developer Intern Apr 2024 — Aug 2024

Roundbytes — Surabaya, Indonesia

- Designed, developed, and maintained client websites using WordPress, applying responsive web design principles to deliver visually consistent digital products.
- Coded responsive, cross-browser web pages using HTML/CSS, TypeScript, and Tailwind CSS across multiple concurrent client projects, refactoring legacy WordPress markup into cleaner, more maintainable component structures.
- Collaborated with the development team to deliver client requirements, participating in sprint planning, code reviews, and coordinated task management across multiple projects.
- Gained practical experience in modern front-end web technologies and deployment practices in a professional agency setting.

RESEARCH & PROJECTS

Undergraduate Thesis — Dental Alveolar Bone Quality Classification 2025 — Present

ITS — Supervised Tugas Akhir Research

- Researching YOLO-based semantic segmentation on CBCT dental images to classify alveolar bone quality under the Lekholm & Zarb D1–D4 grading system.
- Defined the cortical-to-trabecular ratio metric and evaluation framework using DSC, IoU, HD95, Accuracy, Precision, Recall, and F1 Score to analyze model performance.
- Authored a bilingual proposal with Indonesian and English abstracts, documenting the literature review, pseudocode-level methodology, comparative literature gap analysis, and Gantt chart.

NutriWise — Interactive Digital Nutrition Assistant Platform July — Aug 2026

Technology Innovative Challenge (TIC) 9.0 — Universitas Jember

- Lead and developed an interactive health platform combining a health calculator (BMI, daily caloric and fluid needs), a Non-Communicable Disease (NCD) risk screening quiz, and a "Myth vs Fact" nutrition education module to correct common misconceptions.
- Built using React (Vite) and vanilla CSS on the presentation layer, with a hybrid data layer combining browser LocalStorage and cloud storage via Supabase, synchronized through a deep-merge strategy.
- Delivered personalized healthy-lifestyle recommendations based on user calculation results, aligned with the "Good Health and Well-Being" SDG theme.
- Won 1st Place in TIC 9.0, a national web development competition organized by Himpunan Mahasiswa Teknologi Informasi, Universitas Jember, themed "Push the Impact: Web Innovation for Sustainable Development Goals."

PharmaCare+ — AI-Powered Medication Adherence Platform Oct — Dec 2025

Inovasi Surabaya (INOVBOYO) 2025

- Developed a web-based platform that helps elderly users and caregivers improve medication adherence through automated reminders, drug-interaction detection, and online pharmacy consultation services.
- Implemented AI-driven workflows for real-time drug-interaction screening and notifications, then deployed the working web application at pharmacare-gericommunity.lovable.app.
- Competed in the INOVBOYO 2025 regional innovation competition organized by Badan Riset dan Inovasi Daerah Provinsi Jawa Timur, achieving 2nd Runner-Up (Juara Harapan 2).

SkillBridge — Career Matching & Competency Gap Analyzer (Capstone Project) Feb — Jun 2026

PT Sentra Vidya Utama (SEVIMA)

- Built the Laravel 12 + Tailwind CSS v4 front-end for an AI-powered platform that extracts skills from CVs via Claude Sonnet 4.6, analyzes skill gaps against 5 IT role standards, and delivers MCQ-based skill verification and learning path recommendations.
- Implemented the authentication proxy layer (login/register/logout) bridging Laravel sessions with the FastAPI backend's JWT tokens, including custom ApiAuth and ApiGuest middleware for route protection.
- Developed core user-facing Blade views covering the CV upload, skill management, target role selection, skill gap visualization, and learning path pages, consuming a centralized browser-side API client.

TENUNA: The Living Threads — Interactive Editorial Documentary Website June — July 2026

BytesFest 2026

- Built an interactive editorial documentary website preserving and celebrating Sumba Ikat weaving heritage, aligned with SDG 11 (Sustainable Cities and Communities), using React 19, TypeScript, and Vite 8.
- Implemented scroll-driven storytelling with GSAP and ScrollTrigger, synchronizing text transitions, parallax effects, and a custom animated SVG 'Global Thread' element that visually evolves in thickness, color, and pattern as the user scrolls.
- Developed a centralized bilingual (Indonesian/English) system using React Context API, paired with an interactive drag-and-flip language switcher built with Framer Motion for physics-based micro-interactions.

ACHIEVEMENTS

2nd Runner-Up (Juara Harapan 2) — Inovasi Surabaya (INOVBOYO) 2025 Oct — Dec 2025

Badan Riset dan Inovasi Daerah Provinsi Jawa Timur — Regional Competition

- Awarded 2nd Runner-Up in the INOVBOYO 2025 regional innovation competition for PharmaCare+, an AI-powered medication adherence platform for the elderly.

1st Place — Technology Innovative Challenge (TIC) 9.0 July — Aug 2026

Himpunan Mahasiswa Teknologi Informasi, Universitas Jember — National Competition

- Won 1st Place for NutriWise, an interactive digital nutrition assistant platform, under the "Good Health and Well-Being" SDG theme, in a national web development competition themed "Push the Impact: Web Innovation for Sustainable Development Goals."

Semifinalist — Ubaya ILPC 2023 (Programming & Logic Competition) Feb 2023

Universitas Surabaya — Surabaya, Indonesia

- Reached the semifinal stage in a regional competitive programming and logic reasoning competition.

CERTIFICATIONS

- Build an AI Agent — IBM SkillsBuild (AI Agents, IBM watsonx.ai, Agentic AI, Generative AI, Evaluation Metrics, AI Ethics) — 2026
- IT — AI Agent for Programming — IBM SkillsBuild University Education (hosted by Hacktiv8 Indonesia) — 2026
- AWS Academy Cloud Security Foundations — Amazon Web Services (AWS, AWS Cloud, AWS Cloud Security, Cloud Security) — 2026
- Palo Alto Cybersecurity Foundation — Palo Alto Networks

SKILLS

Front-End Development: Vue.js 3, React, Vite, TypeScript, JavaScript, HTML/CSS, Tailwind CSS, Vanilla CSS, shadcn-vue, GSAP, Framer Motion, WordPress, Responsive Web Design

Backend/Database (BaaS): Supabase, RESTful API Integration, PostgreSQL

Programming Languages: Python, JavaScript, TypeScript, SQL, C++

AI / Deep Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, Whisper STT, IBM watsonx.ai, Hugging Face Inference API

Tools & Platforms: Git/GitHub, VS Code, Google Colab, Jupyter Notebook, Supabase

Operating Systems: Windows, Linux (Ubuntu)

Soft Skills: Critical Thinking, Team Collaboration, Time Management, Technical Writing, Problem Solving, Leadership

Languages: Bahasa Indonesia (Native), English — Upper-Intermediate B2